

Lesson **Three**

Software Management & Process

**Effective Project Management:
Traditional, Agile, Extreme**

Ch04: How to Scope a TPM Project

***Presented by
Bowen DU***

Lesson **Three**: Scope a Software Project

Review: Definition of Project Management

Project management is an organized common-sense approach that utilizes the appropriate client involvement in order to meet sponsor needs and deliver the expected incremental business value.

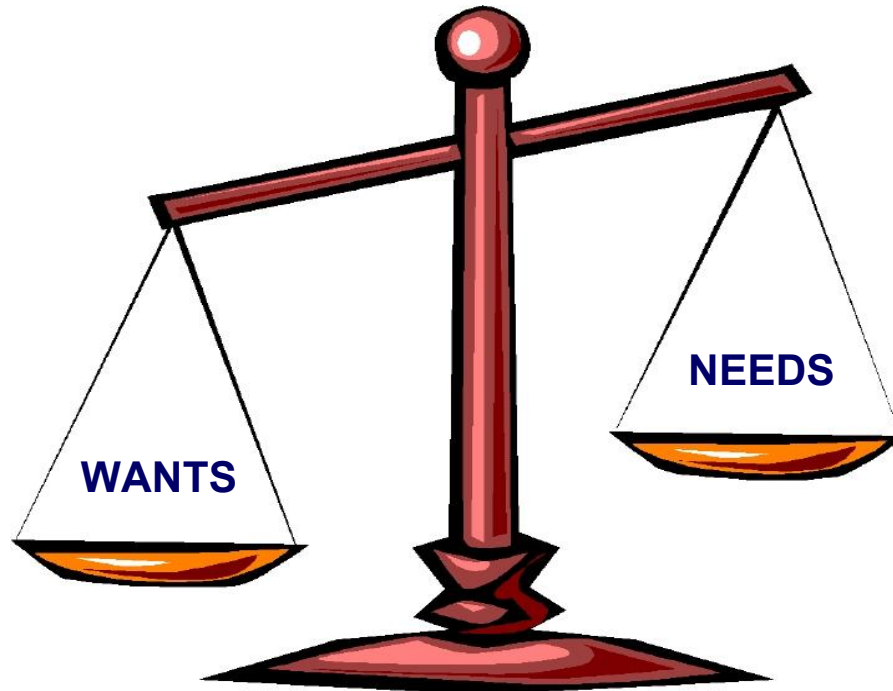
By Wysocki

The application of knowledge, skills, tools and techniques to project activities to meet the project requirements.

By PMI

Lesson **Three**: Scope a Software Project

Client Wants vs. Client Needs Dilemma



What your client wants may not be what your client needs.
Your job is to make sure that what they want is what they need and that you will deliver what they need.

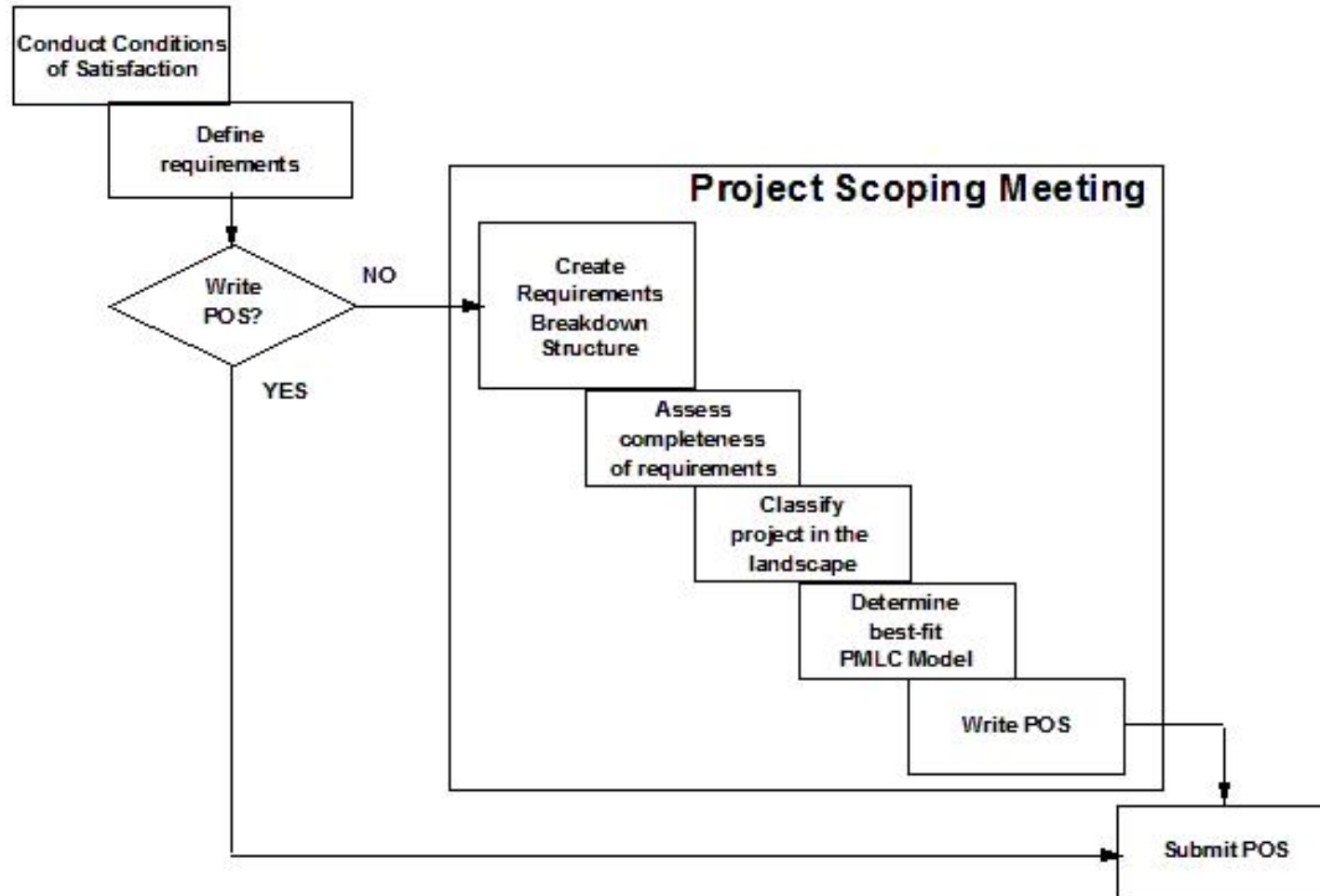
Lesson **Three**: Scope a Software Project

Tips to Managing Client Expectations During Scoping

- Make sure you understand what your client wants/needs/expects
- Make sure the client understands what you will do
- Assure yourself that what your client wants is what your client needs
- Actively include your client in scoping the project
- Put yourself in the shoes of your client
- Meaningfully involve your client wherever possible
- Keep your client informed of project status

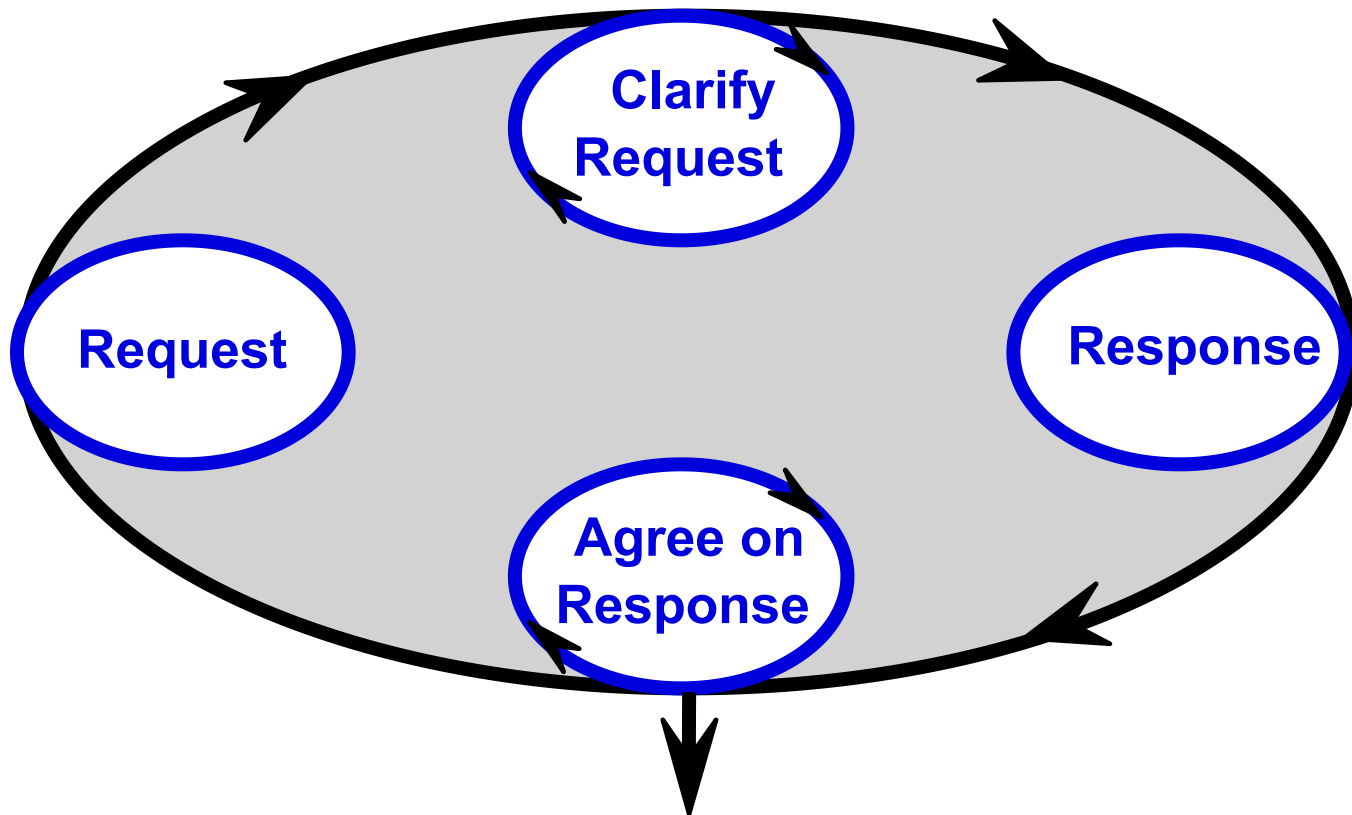
Lesson Three: Scope a Software Project

Project Scoping Process



Lesson **Three**: Scope a Software Project

Establishing Conditions of Satisfaction



**Negotiate agreement and
write Project Overview Statement**

Lesson **Three**: Scope a Software Project

Planning and Conducting the Project Scoping Meeting

- Purpose
 - Document requirements
 - Project Overview Statement
- Attendees
 - Project Manager
 - Client Group
 - Core Team Members
 - The Facilitator & Technographer

Lesson Three: Scope a Software Project

Planning and Conducting the Project Scoping Meeting

- Agenda
 - Introductions
 - Purpose of the Meeting (led by Facilitator)
 - COS (conduct or review if done earlier)
 - Description of current state (led by client representative)
 - Description of problem or business opportunity (led by client representative)
 - Description of end state (led by client representative)
 - Requirements definition and documentation (led by facilitator)
 - Discussion of the gap between current and end state (led by project manager)
 - Choose best fit project management approach to close the gap (led by project manager)
 - Draft and approve the POS (whole scope planning group)
 - Adjourn(休会)

Lesson **Three**: Scope a Software Project

Planning and Conducting the Project Scoping Meeting

- Deliverables
 - RBS creation
 - Assessment of completeness of RBS
 - Project classification
 - Determination of best-fit PMLC Model
 - The POS

Lesson **Three**: Scope a Software Project

Requirement – A Business Value Perspective

Requirement

A requirement is a desired end-state whose successful integration into the solution meets one or more needs and delivers specific, measurable, and incremental business value to the organization.

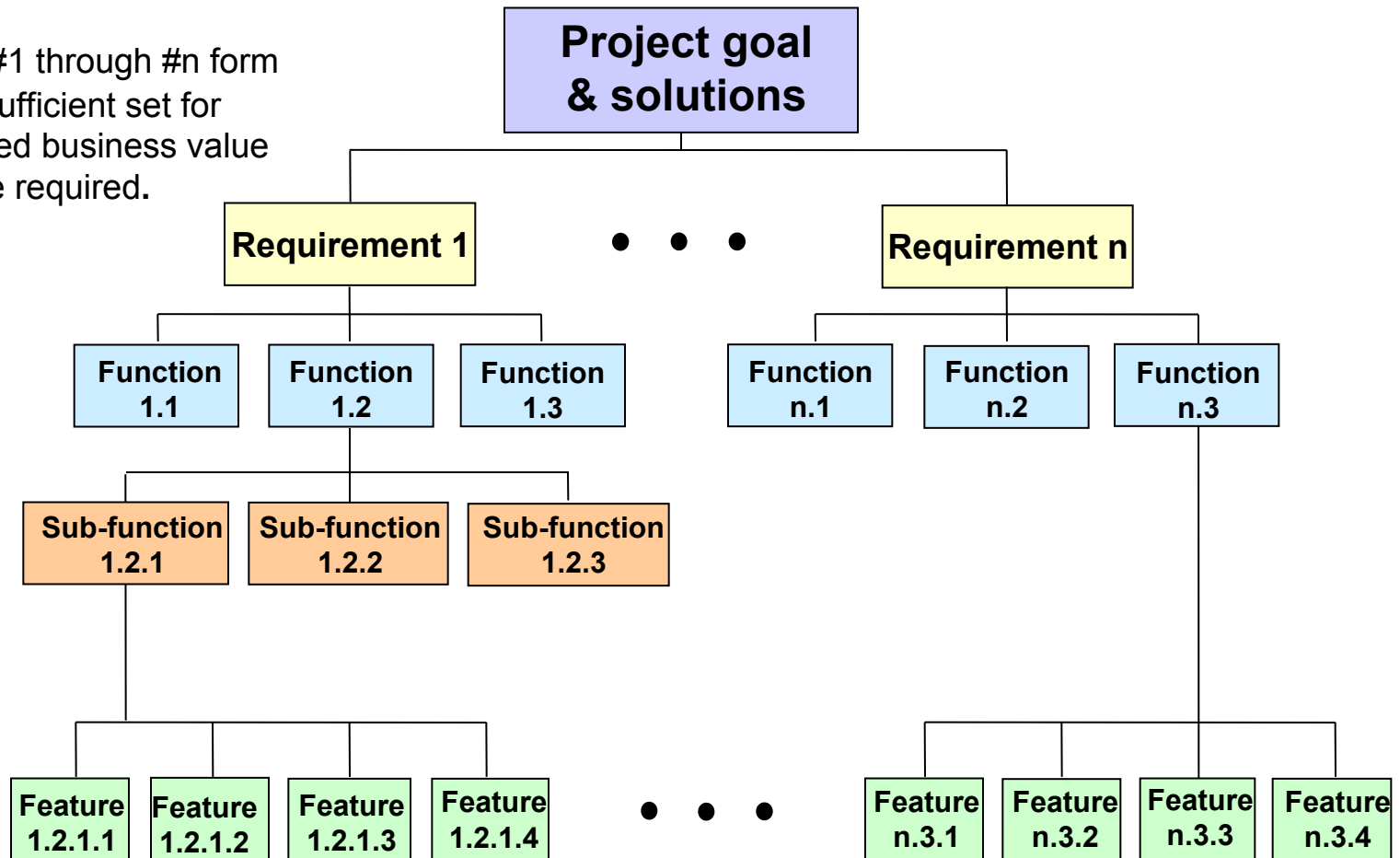
Furthermore, the set of high-level requirements forms a necessary and sufficient set for the attainment of incremental business value.

Lesson **Three**: Scope a Software Project

Building the Requirements Breakdown Structure

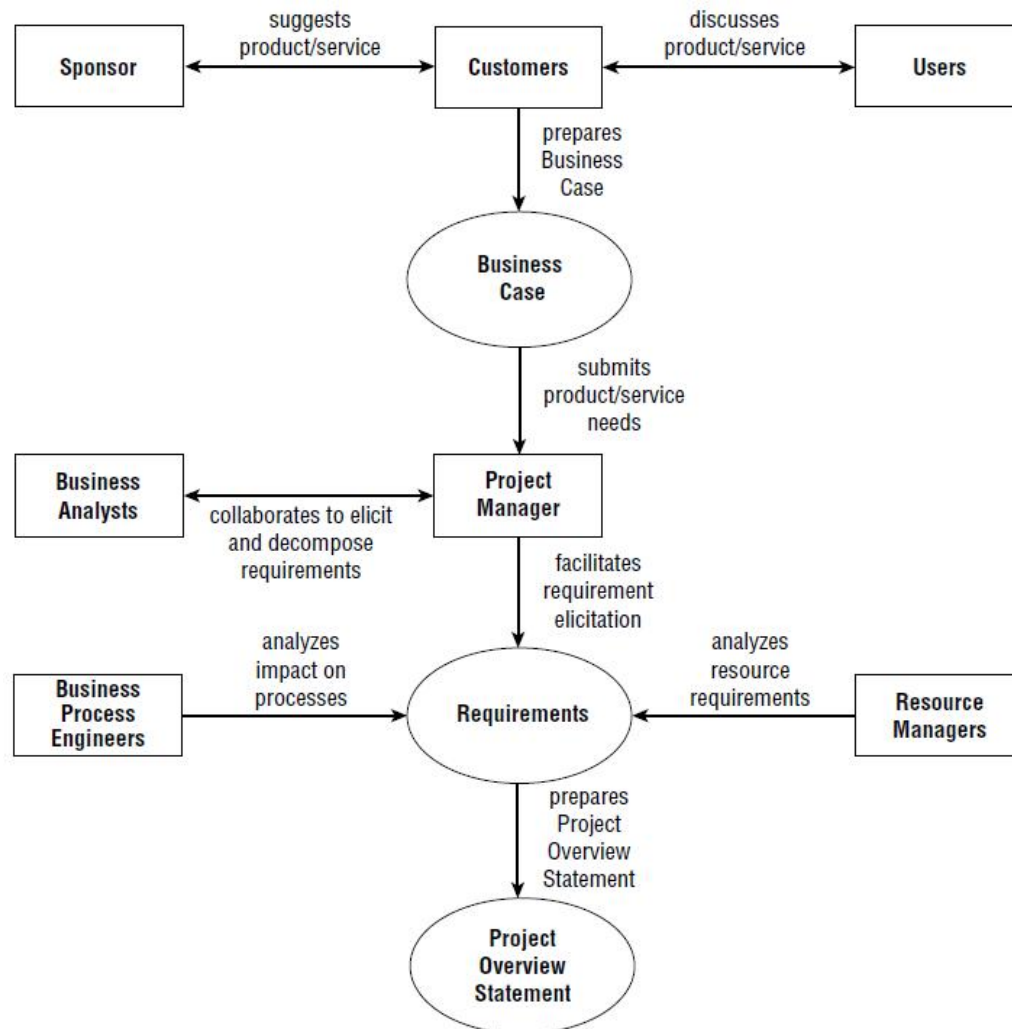
NOTE:

- Requirements #1 through #n form a necessary & sufficient set for attaining expected business value
- Not all levels are required.



Lesson Three: Scope a Software Project

Stakeholder Interaction in Requirements



Lesson Three: Scope a Software Project

Approaches to Requirements Gathering

- Brainstorming
- Facilitated Group Sessions
- Interviews
- Prototyping
- Requirements Workshop

NOTE: These approaches can be used for decomposition too:

- The RBS is most meaningful to the client
- The RBS is deliverables-based
- The RBS is consistent with the PMBOK
- The RBS is client-facing
- The RBS is the higher order part of the WBS

Lesson **Three**: Scope a Software Project

Brainstorming

Strengths

1. Reduces social inhibitions
2. Stimulates idea generation
3. Increases overall creativity of the group

Risks

1. Can be dominated by a strong individual
2. Individual fear of being criticized
3. Evaluation of ideas

Lesson **Three**: Scope a Software Project

Facilitated Group Session Method

Strengths

1. Excellent for cross-functional processes
2. Detailed requirements are documented and verified immediately
3. Resolves issues with an impartial facilitator

Risks

1. Untrained facilitators can lead to negative responses
2. Time and cost of planning and executing can be high

Lesson **Three**: Scope a Software Project

Interview Method

Strengths

1. End-user participation
2. High-level description of functions and processes provided

Risks

1. Descriptions may differ from actual detailed activities
2. Without structure, stakeholders may not know what information to provide
3. Real needs ignored if analyst is prejudiced

Lesson **Three**: Scope a Software Project

Prototyping

Strengths

1. Innovative ideas can be generated
2. Users clarify what they want
3. Users identify requirements that may be missed
4. Client-focused
5. Early proof of concept
6. Stimulates thought process

Risks

1. Client may want to implement prototype
2. Difficult to know when to stop
3. Specialized skills required
4. Absence of documentation

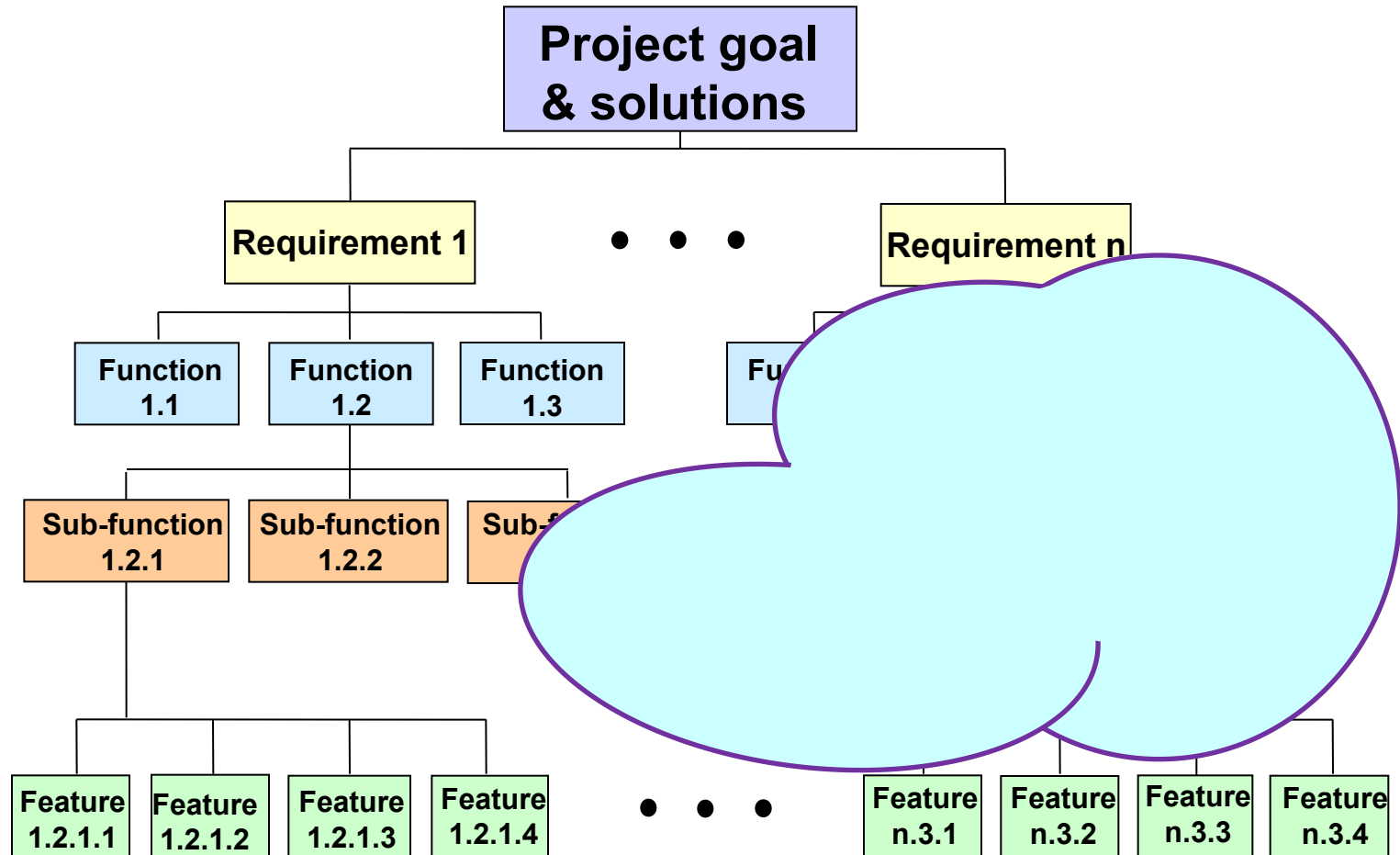
Lesson Three: Scope a Software Project

Categories of Requirements

- **Functional**
 - Functional requirements specify what the product or service must do.
- **Non-functional**
 - Non-functional requirements demonstrate properties that the product or service should have in order to do what must be done.
- **Product/project constraints**
 - Product/project constraints are those requirements that, on the surface, resemble design constraints or project constraints.

Lesson Three: Scope a Software Project

RBS – The Reality



Lesson **Three**: Scope a Software Project

Characteristics of the RBS

- The RBS is intuitive and most meaningful to the client.
- The RBS is a deliverables based approach.
- The RBS is consistent with the PMI PMBOK.
- The RBS remains client facing as long as possible into the planning exercise.

Lesson Three: Scope a Software Project

Advantages of using the RBS

- Does not require a trained facilitator
- Does not require learning one of the contemporary approaches to requirements gathering
- Presents an intuitive approach to gathering requirements
- Allows the client to work with the project team in an environment familiar to them, i.e., to stay in their own comfort zone
- Paints a clear picture of the degree to which the solution is clearly defined
- Provides the input needed to choose the best-fit PMLC Model

Lesson **Three**: Scope a Software Project

The Challenge of Requirements Management

- Many different types of requirements at different levels of detail.
- Requirements are not independent of one another and may create conflict situations.
- Many interested and responsible parties with different needs.
- Change as a result of changing business conditions.
- Can be time-sensitive.
- Shuttle diplomacy to resolve differences.
- Assess completeness of requirement decomposition.

Lesson **Three**: Scope a Software Project

Hints in Deciding which PMLC Model to Use

- The degree to which the RBS is complete is the major factor in deciding which PMLC Model to use.
- Consider using the highest level of decomposition in the Objective section of the POS and leaving creation of the RBS and WBS for the Planning Phase.
- The highest level requirements could be those that deliver business value. Senior managers might prefer this.

Lesson **Three**: Scope a Software Project

When to Use each PMLC Model

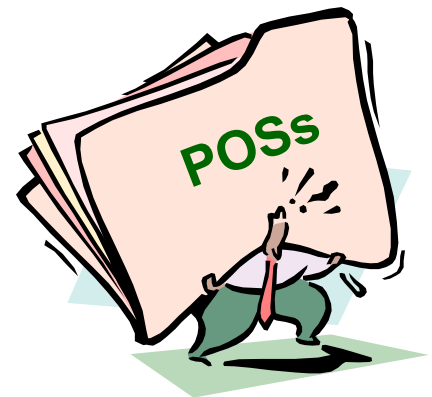
PMLC Model Type	When to Use It
Linear	The solution and requirements are clearly defined.
	You do not expect too many scope change requests.
	The project is routine and repetitive.
	You can use established templates.
Incremental	Same conditions as the Linear approach, but the client wants to deploy business value incrementally.
	There may be some likelihood of scope change requests.
Iterative	You feel that requirements are not complete or may change.
	You will learn about remaining requirements in the course of doing the project.
	Some features of the solution are not yet identified.
Adaptive	The solution and requirements are only partially known.
	There may be functionality that is not yet identified.
	There will be a number of scope changes from the client.
	The project is oriented to new product development or process improvement.
	The development schedule is tight and you can't afford rework or re-planning.
Extreme	The goal and solution are not clearly known.
	The project is an R & D type project.

Lesson Three: Scope a Software Project

Purpose of the Project Overview Statement (POS)

A one-page description that is:

- A general statement of the project.
- A reference for the planning team.
- A decision aid for the project.
- To get management approval to plan the project.



Lesson **Three**: Scope a Software Project

Contents of the Project Overview Statement (POS)

Problem/Opportunity

Project Goal

Project Objectives

Success Criteria

Assumptions/Risks/Obstacles

Lesson Three: Scope a Software Project

Example POS



PROJECT OVERVIEW STATEMENT	Project Name	Office Supply Cost Reduction	Project No.		Project Manager	PAUL BEARER
	Problem/Opportunity					
	Our cost reduction task force reports that office supply expenses have exceeded budget by an average of 4% for each of the last three fiscal years. In addition an across the board budget cut of 2% has been announced and there is an inflation rate of 3% estimated for the year.					
	Goal					
	To implement a cost containment program that will result in office supply expenses being within budget by the end of the next fiscal year.					
Objectives						
1. Establish a departmental office supply budgeting and control system. 2. Implement a central stores for office and copying supplies. 3. Standardize the types and brands of office supplies used by the company. 4. Increase employee awareness of copying practices that can reduce the cost of meeting their copying needs.						
Success Criteria						
1. The total project cost is less than 4% of the current year office supply budget. 2. At least 98% of office supply requests are filled on demand. 3. At least 90% of the departments have office supply expenses within budget. 4. No department office supply expense exceeds budget by more than 4%.						
Assumptions, Risks, Obstacles						
1. Central stores can be operated at or below the breakeven point. 2. Users will be sensitive to and supportive of the cost containment initiatives. 3. Equitable office supply budgets can be established. 4. Management will be supportive and consistent. 5. The existing inventory control system can support the central stores operation.						
Prepared By		Date		Approved By		Date
Olive Branch		9/2/04		Del E. Lama		9/3/04

Lesson **Three**: Scope a Software Project

POS Problem/Opportunity

- A problem needing resolution or an untapped business opportunity.
- A statement of fact that everyone would agree to. It stands on its own.
- This is the foundation on which the proposed project will be based.

Lesson Three: Scope a Software Project

POS Project Goal

- A one or two sentence statement of how you intend to address the stated problem/opportunity.
- A scoping statement that bounds the project you are proposing.

Lesson **Three**: Scope a Software Project

POS Project Objectives

- 5 or 6 brief statements that further bound your project goal.
- From these statements it is clear what is in the proposed project.
- These statements might identify major project deliverables.
- These statements form a necessary and sufficient set of objectives.

Under what conditions might you consider using the high-level requirements as the project objectives?

Lesson **Three**: Scope a Software Project

POS Project Success Criteria

IRACIS

IR	Increasing Revenue
AC	Avoiding Cost
IS	Improving Service



Use quantitative metrics only!
How much and by when?

Lesson Three: Scope a Software Project

POS Assumptions, Risks and Obstacles

- Technological
 - New to the company.
 - Obsolescence.
- Environmental
 - Management change.
 - Staff turnover.
- Interpersonal
 - Working relationships.
- Cultural
 - Fit to the company.
- Causal Relationships
 - Will the solution solve the problem.

Lesson **Three**: Scope a Software Project

POS Attachments

- Risk Analysis
- Financial Analyses
 - Feasibility studies
 - Cost/benefit analysis
 - Breakeven analysis
 - Return on investment

Lesson Three: Scope a Software Project

Gaining Approval to Plan the Project

- Expected Review Questions from Management
 - How important is the problem or opportunity to the organization?
 - How is the project related to our CSFs?
 - Does the goal statement related directly to the problem or opportunity?
 - Are the objectives clear representations of the goal statement?
 - Is there sufficient business value as measured by the success criteria to warrant further expenditures on this project?
 - Is the relationship between the project objectives and the success criteria clearly established?
 - Are the risks too high and the business value too low?
 - Can senior management mitigate the identified risks?

Lesson **Three**: Scope a Software Project

Participants in the Approval Process

- Core project team.
- Project team.
- Project manager.
- Resource managers.
- Function/process managers.
- Clients.
- Senior management.

Lesson **Three**: Scope a Software Project

Tools, Templates, & Processes Used to Scope a Project

- Conditions of Satisfaction
- Project Scoping Meeting
- Requirements Elicitation
- Brainstorming
- Facilitated Group Sessions
- Interviews
- Prototyping
- Requirements Workshops
- Project Overview Statement
- Approval to Plan the Project

Lesson **Three**: Scope a Software Project

Summary of Lesson **3**

- Using tools, templates, processes to scope a software project
- Managing client expectations
- Project scoping process
- Project scoping meeting & deliverables
- Requirements breakdown structure (RBS)
- Project classification
- Choosing the best-fit PMLC Model
- Project Overview Statement

Lesson **Three**: Scope a Software Project

Assignment of Lesson **3**

- Team assignment: POS of course project
 - Analysis and design your course project, write your POS.
 - The Deadline: Before **23:59** on **March 22**, 2026.
 - Team leader should submit the POS in the form of replying to the homework at Canvas.